

Applied AI Portfolio Site Content

Recruiter-facing copy and structure for Nicholas Nelson's portfolio site

1. Core Site Identity

This portfolio should present Nicholas Nelson as an early-career applied AI builder with a clear focus on practical AI systems, context engineering, retrieval-based tools, evaluation design, technical communication, and polished project work.

Primary Identity

Nicholas Nelson
Applied AI Systems Student

Recommended Site Title

Nicholas Nelson | Applied AI Systems Portfolio

Main Tagline

Turning complex AI ideas into clear, testable systems.

Supporting Positioning Line

I am an IT student focused on applied AI, context engineering, technical communication, and building practical digital tools that turn messy ideas into organized systems.

2. Hero Section

The hero section should immediately make the professional direction clear. It should feel polished, direct, and hireable without pretending to be a senior-level AI architect.

Nicholas Nelson
Applied AI Systems Student

Turning complex AI ideas into clear, testable systems.

I am an IT student focused on applied AI, context engineering, technical communication, and building practical digital tools that turn messy ideas into organized systems.

Hero Buttons

- View Projects
- GitHub

- LinkedIn

3. About Section

I am an Information Technology student at New Jersey Institute of Technology with experience across web development, data analysis, applied AI projects, and technical project building. My work has included generative AI output optimization, MySQL database design, machine learning experiments, mobile app development, and full-stack web applications.

I am drawn to applied AI because I like the space between ideas and implementation. I like figuring out what a tool should do, what information it needs, how the output should be evaluated, and how to explain the system clearly to someone else. My goal is to keep building projects that show technical growth, practical thinking, and strong communication.

4. Career Direction

My current professional direction is applied AI engineering and AI systems design. I am interested in roles where technical builders help turn AI ideas into working products, evaluate whether AI systems perform well, and communicate complex technical decisions clearly.

The Applied AI Architect path stood out to me because it combines software engineering, AI product thinking, technical communication, evaluation frameworks, and architecture design. It is not only about using AI. It is about understanding what a product needs, shaping the right context, testing outputs, and building systems that people can actually rely on.

5. Professional Focus Areas

Applied AI Systems

I am interested in building AI tools that are useful, structured, and grounded in real user needs. My work focuses on turning broad ideas into systems with clear inputs, outputs, and evaluation points.

Context Engineering

I care about how AI systems receive instructions, source material, constraints, tone, and user goals. Strong AI output depends on strong context, not just a clever prompt.

Evaluation and Testing

I am interested in testing whether AI outputs are accurate, useful, consistent, and aligned with the task. I want my AI projects to be checked and validated, not just impressive at first glance.

Technical Communication

I enjoy making technical ideas easier to understand. Whether I am documenting a project, presenting a system, or explaining a design choice, I want the reasoning behind the work to be clear.

6. Featured Project: StoryLens AI

Project Name

StoryLens AI

One-Sentence Pitch

StoryLens AI helps writers turn scattered story notes into source-grounded scene plans by retrieving relevant lore, generating structured storyboards, and checking continuity.

Short Project Card

A RAG-style storyboarding tool that retrieves story context, creates structured scene plans, and checks continuity against the source material.

Full Project Description

StoryLens AI is a source-grounded storyboarding assistant for writers building long-form stories with complex lore, character arcs, and style rules. The app retrieves relevant story notes, character details, worldbuilding rules, previous chapter summaries, and style guidelines before generating a structured scene plan.

Instead of acting like a generic AI writing assistant, StoryLens AI is designed around continuity. It helps writers plan scenes while keeping the story grounded in the material they already created. The app also evaluates the storyboard for source alignment, character consistency, worldbuilding consistency, tone fit, style match, scene clarity, and hallucination risk.

Why I Built It

I built StoryLens AI around a real problem in AI-supported creative planning: long-form stories require memory, consistency, and style control, but general AI chats can lose track of important details over time. This project explores how retrieval-based context and structured evaluation can make AI-supported storyboarding more reliable and useful.

Core Features

- Story knowledge base for character notes, lore, plot points, style rules, and previous chapter summaries
- Scene or chapter request input
- Retrieval panel showing the source material used to shape the storyboard
- Structured storyboard output with scene goal, opening situation, key beats, conflict, continuity notes, and ending hook
- Continuity evaluation across source alignment, character consistency, worldbuilding consistency, tone fit, style match, scene clarity, and hallucination risk
- Copy or export storyboard as Markdown

Skills Demonstrated

- Next.js
- TypeScript
- Zod
- Vitest
- RAG-style retrieval
- Context engineering
- Evaluation design
- Source-grounded generation
- Technical writing
- Product thinking

7. Other Featured Projects

Revised Midterm Project

This revised midterm project was improved for portfolio presentation through clearer writing, stronger organization, cleaner visual design, and a more professional explanation of the project purpose. The revision focused on making the work easier to understand as a finished portfolio piece rather than leaving it as a class submission.

TODO: Add project title, topic, tools used, specific improvements, live link, GitHub link, and screenshots if available.

UNSW-NB15 Intrusion Detection

A machine learning project that classifies network traffic as normal or attack using supervised and unsupervised models. This project used models including Logistic Regression, Decision Tree, Random Forest, and Isolation Forest, along with an end-to-end scikit-learn pipeline for preprocessing, imbalance handling, and hyperparameter optimization.

- Skills: Python, Pandas, scikit-learn, machine learning, data analysis, intrusion detection, model evaluation

Taro App

A chore management app with authentication, task tracking, rewards, groups, leaderboards, and statistics. The project demonstrates practical product thinking through user accounts, shared progress, engagement mechanics, and team-based interaction.

- Skills: Kotlin, Android Studio, Firebase, product design, mobile development, Agile

Sakila Movies

A full-stack film rental platform with search, detailed views, RESTful APIs, and customer management. The project demonstrates full-stack application development using React, Node.js, Express, and MySQL.

- Skills: React, Node.js, Express, MySQL, REST APIs, CRUD, full-stack development

8. Skills and Tools Section

Applied AI and Evaluation

- Prompt engineering
- Context engineering
- RAG-style retrieval
- Evaluation design
- Output analysis
- Source-grounded generation

Software Development

- HTML
- CSS
- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express
- PHP
- Laravel
- Kotlin

Data and Machine Learning

- Python
- Pandas
- NumPy
- scikit-learn
- Machine learning pipelines
- Data preprocessing

- Model evaluation
- MySQL

Tools and Professional Strengths

- Git
- GitHub
- Docker
- Android Studio
- Zod
- Vitest
- Technical communication
- Project planning
- Documentation
- Peer mentoring
- Presentation

9. Visual Direction

The site should feel like a modern AI research-lab portfolio: polished, readable, structured, and technical without being cold. It should avoid cyberpunk overload, fake hacker styling, and generic buzzwords.

Recommended Style

- Dark navy or charcoal background
- Off-white body text
- Blue, cyan, or soft violet accents
- Clean project cards
- Strong spacing and readable typography
- Subtle technical visual details, not distracting effects

10. Recommended Page Structure

- Hero
- About
- Career Direction
- Professional Focus Areas

- Featured Projects
- Skills and Tools
- Contact and Links

11. Contact Section

Use this section near the bottom of the site and make the same links visible in the hero or navigation.

Let's connect.

I am building toward applied AI and AI systems work through practical projects, technical writing, and structured product thinking.

GitHub: <https://github.com/Eagolo135>

LinkedIn: <https://www.linkedin.com/in/nicholas-nelson-21b5a329b/>

Email: nan@njit.edu

This document contains only recruiter-facing portfolio site content and public project positioning.